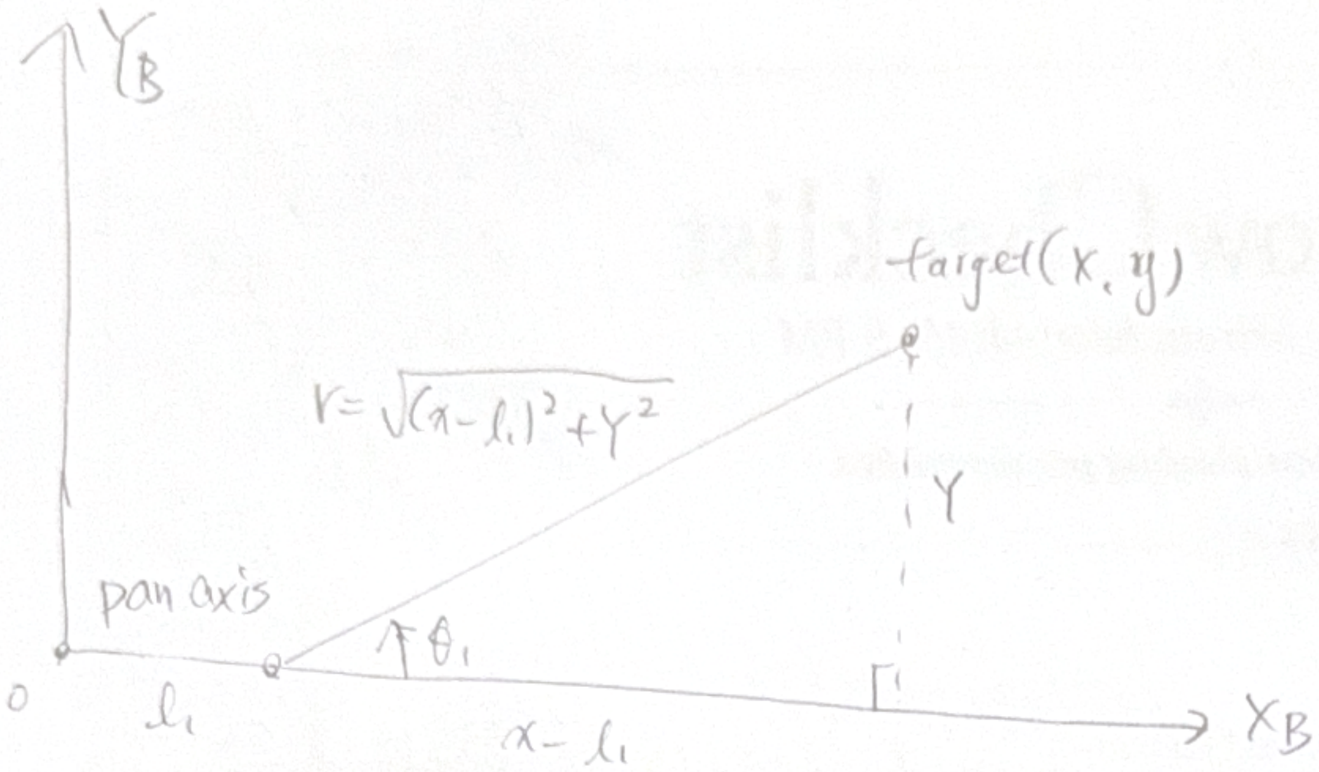
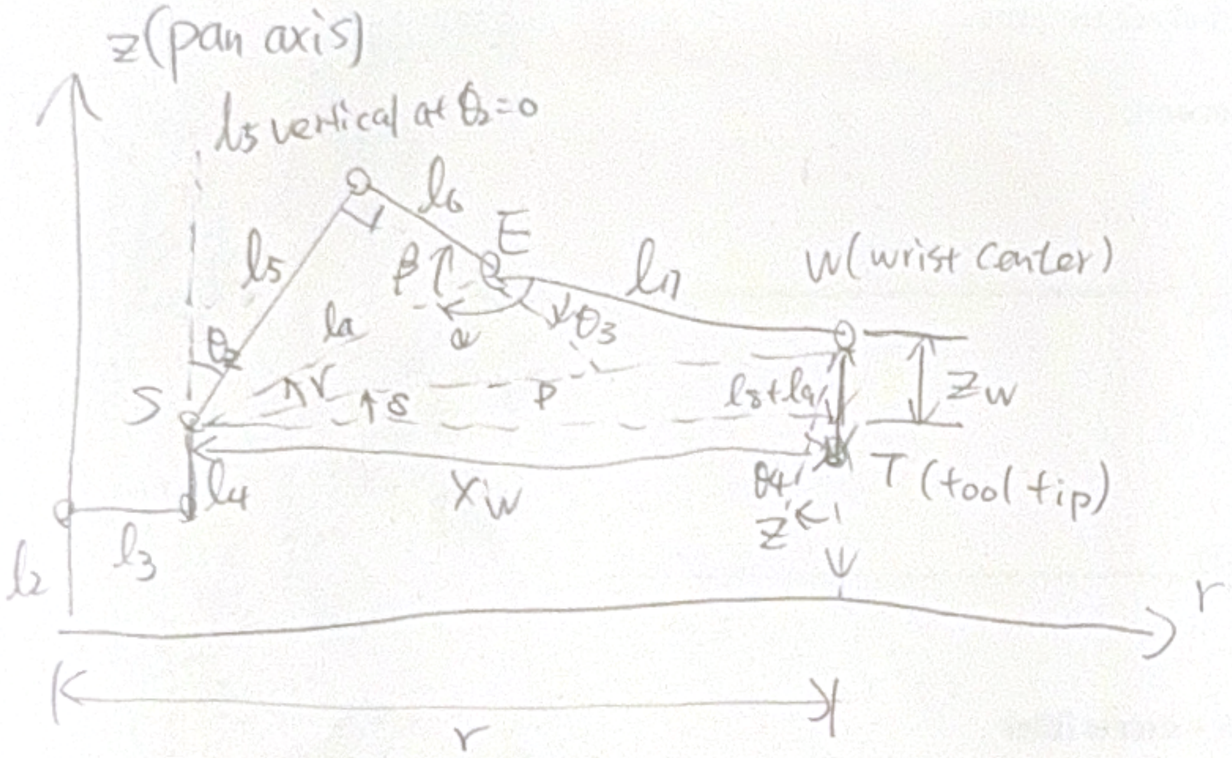




- Constants: $l_a=\sqrt{(l_5^2+l_6^2)}$, $\beta=\text{atan2}(l_5, l_6)$
- Intermediate Quantities (from the target X, Y, Z)

$r=\sqrt{(x-l_1)^2+Y^2}$

$x_w=r-l_3$

$z_w=z+l_8+l_9-(l_2+l_4)$

$P=\sqrt{x_w^2+z_w^2}$

$\delta=\text{atan2}(z_w, x_w)$

$\cos\alpha=\dfrac{(l_a^2+l_7^2-P^2)}{2\cdot l_a\cdot l_7}$

$\cos\gamma=\dfrac{(l_a^2+P^2-l_7^2)}{2\cdot l_a\cdot P}$

- joint angles

$\theta_1=\text{atan2}(Y, x-l_1)$

$\theta_2=\beta-\gamma-\delta$

$\theta_3=\pi-\beta-\alpha$

$\theta_4=-(\theta_2+\theta_3)$

$\theta_5=0$

Section 8.

J is a 3x4 matrix that maps the joint rates $(\dot{q}_1, \dot{q}_2, \dot{q}_3, \dot{q}_4)$ to the tool-tip velocity $(\dot{x}, \dot{Y}, \dot{z})$ expressed in the base frame.

$A = q_2$, $B = q_2 + q_3$, $C = q_2 + q_3 + q_4$

$r = l_3 + l_5 \sin A + l_6 \cos A + l_7 \cos B + (l_8 + l_9) \cos C$

$Z = (l_2 + l_4) + l_5 \cos A - l_6 \sin A - l_7 \sin B - (l_8 + l_9) \sin C$

$x = l_1 + r \cdot \cos q_1$

$Y = -r \cdot \sin q_1$

$Z = z$

$r_{q2} = l_5 \cos A - l_6 \sin A - l_7 \sin B - (l_8 + l_9) \sin C$

$r_{q3} = -l_7 \sin B - (l_8 + l_9) \sin C$

$r_{q4} = -(l_8 + l_9) \sin C$

$\dot{z}_{q2} = -l_5 \sin A - l_6 \cos A - l_7 \cos B - (l_8 + l_9) \cos C$

$z_{q3} = -l_7 \cos B - (l_8 + l_9) \cos C$

$z_{q4} = -(l_8 + l_9) \cos C$

$$J = \begin{bmatrix} -r \sin q_1 & \cos q_1 r_{q2} & \cos q_1 r_{q3} & \cos q_1 r_{q4} \\ -r \cos q_1 & -\sin q_1 r_{q2} & -\sin q_1 r_{q3} & -\sin q_1 r_{q4} \\ 0 & z_{q2} & z_{q3} & z_{q4} \end{bmatrix}$$